OPSC7312 POE Part 1 – Planning and Design

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Bachelor of Computer and information Sciences in Application Development

OPSC7312

25 August 2023

Reece Wanvig

I here declare that I did not plagiarize the content of this Assignment and that this is my own work.

Assignment submitted via SafeAssign or Turnitin: ☑

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Part 1 — Planning and Design

Introduction

The CRZ bird watch app project aims to bridge the gap between urban living and the appreciation of nature's beauty, specifically focusing on birdwatching enthusiasts. The CRZ bird watch app seeks to provide a comprehensive solution to these challenges by offering an intuitive, user-friendly platform that locates birdwatching hotspots and provides precise directions. The primary objective of this project is a fully functional mobile application available on a major platform (Android) with an intuitive user interface.

Overview

Name:

'CRZ – Birdwatch™'

Icon:

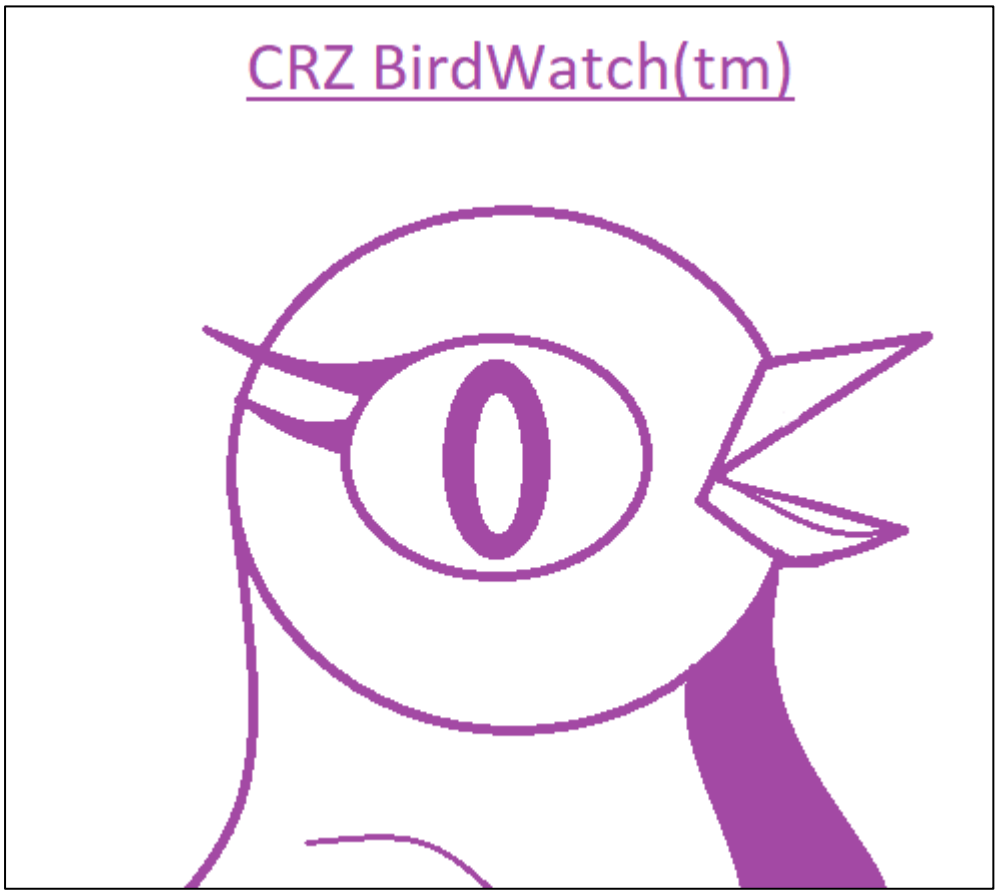


Figure 1

Innovative Features Descriptions:

1)Nearby Feature: These features with help Users to seek out areas where birds can be found more easily, this will use the User's current location to pinpoint areas near that are ideal for birdwatching, (Birda, 2023).

2)Distance-tracking Feature: An extension of the Nearby feature, helping calculate the current position of the Users to provide travel-relevant information. The current position is stored, the desired location stored, then the application will calculate the distance between these two points before presenting it the user (eBird by Cornell Lab, 2023).

3)Checklists Feature: Allows Users to record and submit information regarding birds that they encounter. This helps Users record information and store it on the system effectively, data input into the checklist will be stored as a series of variables underneath one specific checklist slot (GoBird, – Guide to Nearby Birds, 2023).

The API we want to use:

We will be using eBird API 2.0.

The main reason for this choice is due to the fact it is the stock choice for our OPSC7312 POE Part 1 question paper. This option seems the most reasonable, since other options we explored were unreliable.

Upon further research, we are also using it due to the fact that it is a replacement for 1.1 APIs, considering these are now retired (Documenter.getpostman.com, 2023). This Pai will provide a set of wrapper functions for accessing the endpoints in the eBird API 2.0 (Pypi.org, 2023) system.

Detailed list of the requirements

These details were gained from the IIE, (2023) POE question paper.

Minimum Requirements for the application:

1)'The user must be able to register using the app' (IIE, 2023: 4):

Users will be able to create an account on the program, in order to store their personal data regarding the App. The User will be able to store their 'Name', 'Email Address' and 'Password' (Birda, 2023: 4).

2)'The user must be able to login to the app using a username and password' (IIE, 2023: 4):

Username and Password will be stored in the application's database, under specific a User (Birda, 2023).

3)'The user must be able to change their settings including but not limited to the following' (IIE, 2023: 4):

- **'Toggle between using the metric system (kilometres) or using the imperial system (miles) (IIE, 2023: 4):'**

The User will have to be able to alter the units of measurement they want to use. The distances displayed by the system will be manually changed in the 'setting' page, works as a drop down-menu (eBird by Cornell Lab, 2023).

- **'The maximum distance that the user is willing to travel to a bird watching hotspot (IIE, 2023: 4)':**

The maximum distance will have to be calculated by the App and then presented to the User graphically. This maximum distance can be manipulated by the user in the Nearby features web page (GoBird – Guide to Nearby Birds, 2023).

4)'The user must be able to view nearby birding hotspots on a map (use a webservice to obtain this information…The map component must be embedded in the app' (IIE, 2023: 4):

Hotspots will have to be identified by the App, stored in it's database and then stored as viewable data (eBird by Cornell Lab, 2023).

5)'The app must display hotspots filtered according to the preferred maximum distance chosen by the user' (IIE, 2023: 4):

A function to filter the hotspots stored in the App via distance will be needed (eBird by Cornell Lab, 2023).

6)'The app must display the user's current position on the map' (IIE, 2023: 4):

A location hosting function will have to be included. There may be a way to use the 'GoogleMaps' (GoogleMaps, 2023). Program found on most android devices to immediately attain, stores and show the user's position.

7)'The user must be able to select a hotspot…to get directions to the hotspot' (IIE, 2023: 4):

The App must be able to route roots and store this data, when once hotspot is selected the User will be shown the route as well as the distance (Birda, 2023).

8)'The app must then be able to calculate the best route between your current location and the hotspot. (…create this algorithm yourself or obtain this information from a webservice.)' (IIE, 2023: 4):

Route calculation must be accurate and easily shown. It will be displayed in a text box (GoBird Guide to Nearby Birds, 2023).

9)'The app must display this route visually (on…embedded map)' (IIE, 2023: 4):.

The App will have a map feature to visually indicate this data. It's best to highlight the route with a brightly coloured line (GoBird – Guide to Nearby Birds, 2023).

10)'…must allow the user to save a bird observation at their currently location' (IIE, 2023: 4):

This will be included in the checklist function in order to store the data in an orderly manner (GoBird – Guide to Nearby Birds, 2023).

11)'…must allow the user to view all their bird observations' (IIE, 2023: 4):.

This data will be stored in a checklist. Users will be able to jot down their information into a checklist on the app (GoBird – Guide to Nearby Birds, 2023).

12)'The registration information must be stored in an online, hosted authentication service. (Part 3…' (IIE, 2023: 4):

This feature will be assessed and then implemented closer to the due date. For the first implementation, the information will just be stored on the App itself (Birda, 2023).

13)'The settings and user bird observations must be stored in an online, hosted database. (Part 3…' (IIE, 2023: 4):

This feature will be assessed and then implemented closer to the due date. For the first implementation, the information will just be stored on the App itself (Birda, 2023).

14)'The app must display bird observations on the map. (Part 3…' (IIE, 2023: 4):

This feature will be assessed and then implemented closer to the due date The IIE (2023) (GoBird – Guide to Nearby Birds, 2023).

Our own features to be added:

1)Nearby Feature: These features with help Users to seek out areas where birds can be found more easily. This was taken from Birda (2023), due to the practicality.

2)Distance-tracking Feature: This is an extension of the Nearby feature, helping to calculate the current position of the Users to provide travel-relevant information. This information will be conveyed visually, it will also adjust according to a chosen unit of measurement. This was taken from eBird by Cornell Lab (2023), this is a convenient feature to add.

3)Checklists Feature: Allows Users to record and submit information regarding birds that they encounter. This will help Users record information and have it be stored on the system effectively. This was taken from GoBird – Guide to Nearby Birds (2023), this was an intuitive method of data capturing all Users could understand.

User interface design

Screenshots:

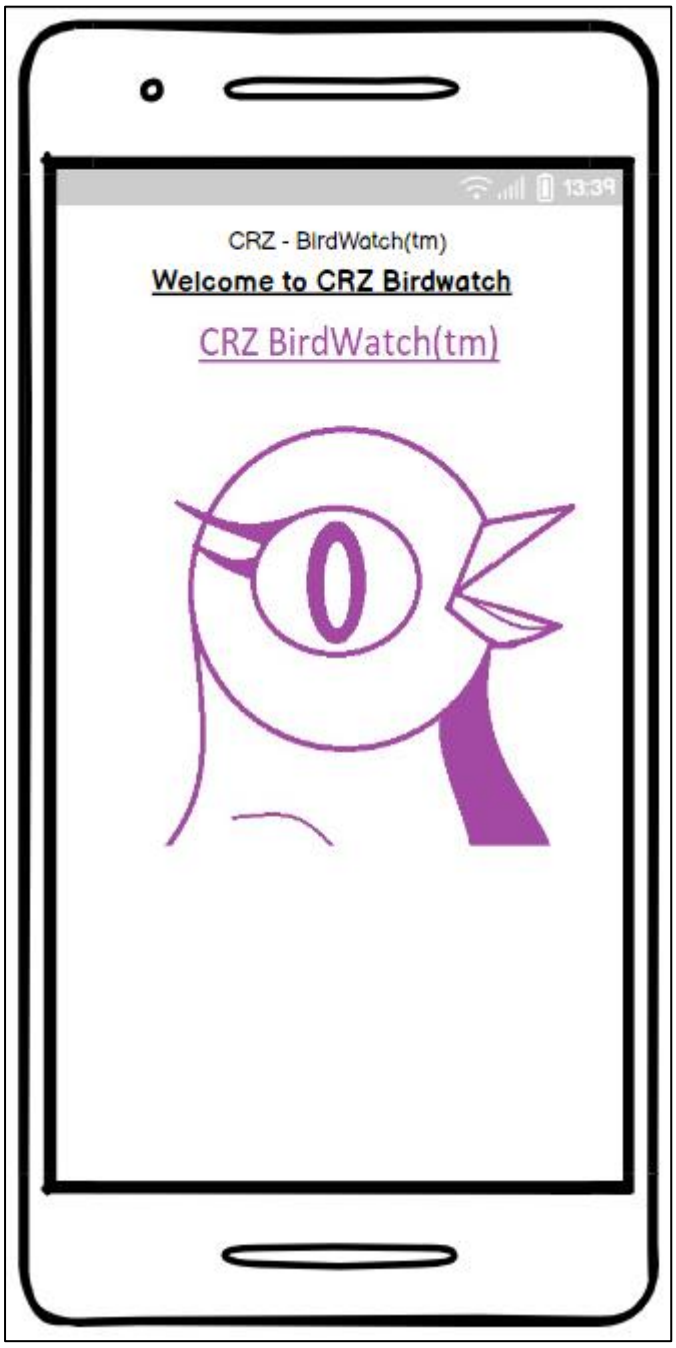


Figure 2

Opening Page

This is the page that will appear as the App starts up.

This page won't be able to be manipulated in any way, it's basically a splash screen.

This practice was observed in Birda (2023). This was added for the sake of good branding.

General practices in Android App development will be carried out using notes from Ratabouil, S (2015), specifically their guide on 'Android NDK'.

The approach to UI design was inspired by Wilson, J. (2016)'s techniques for UI.

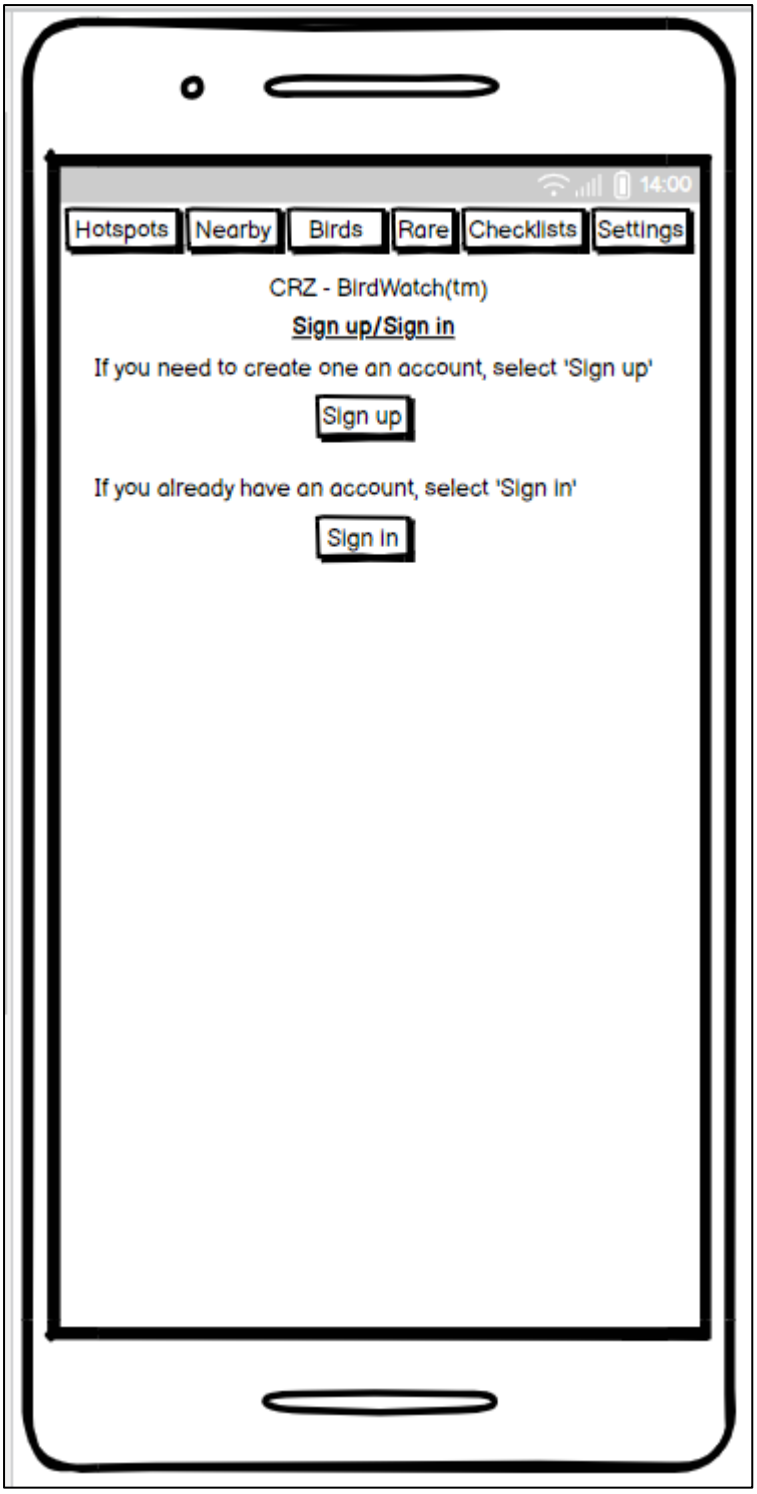


Figure 3

Sign up/Sign in Page

The web page dedicated to helping Users create an account or access their old one.

This is a common feature of most apps (Birda, 2023).

*'Sign up' to create an account on the app.

*'Sign in' to access their account on the app.

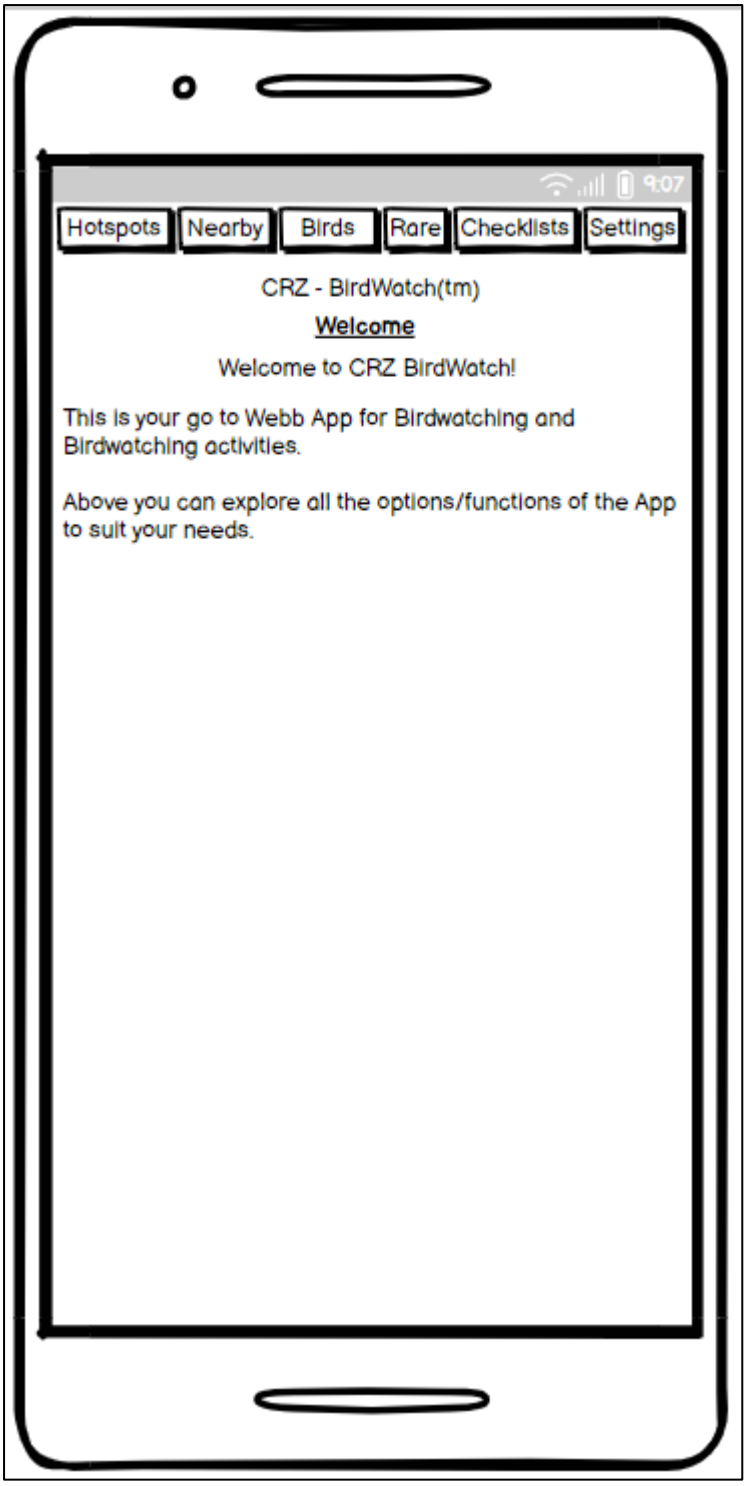


Figure 4

Welcome Page

*"Welcome to CRZ BirdWatch!"

*"This is your go to Webb App for Birdwatching and Birdwatching activities…Above you can explore all the options/functions of the App to suit your needs."

The purpose of the application is explained, as is the different functionalities that it can provide to the User.

This feature was included in Birda (2023).

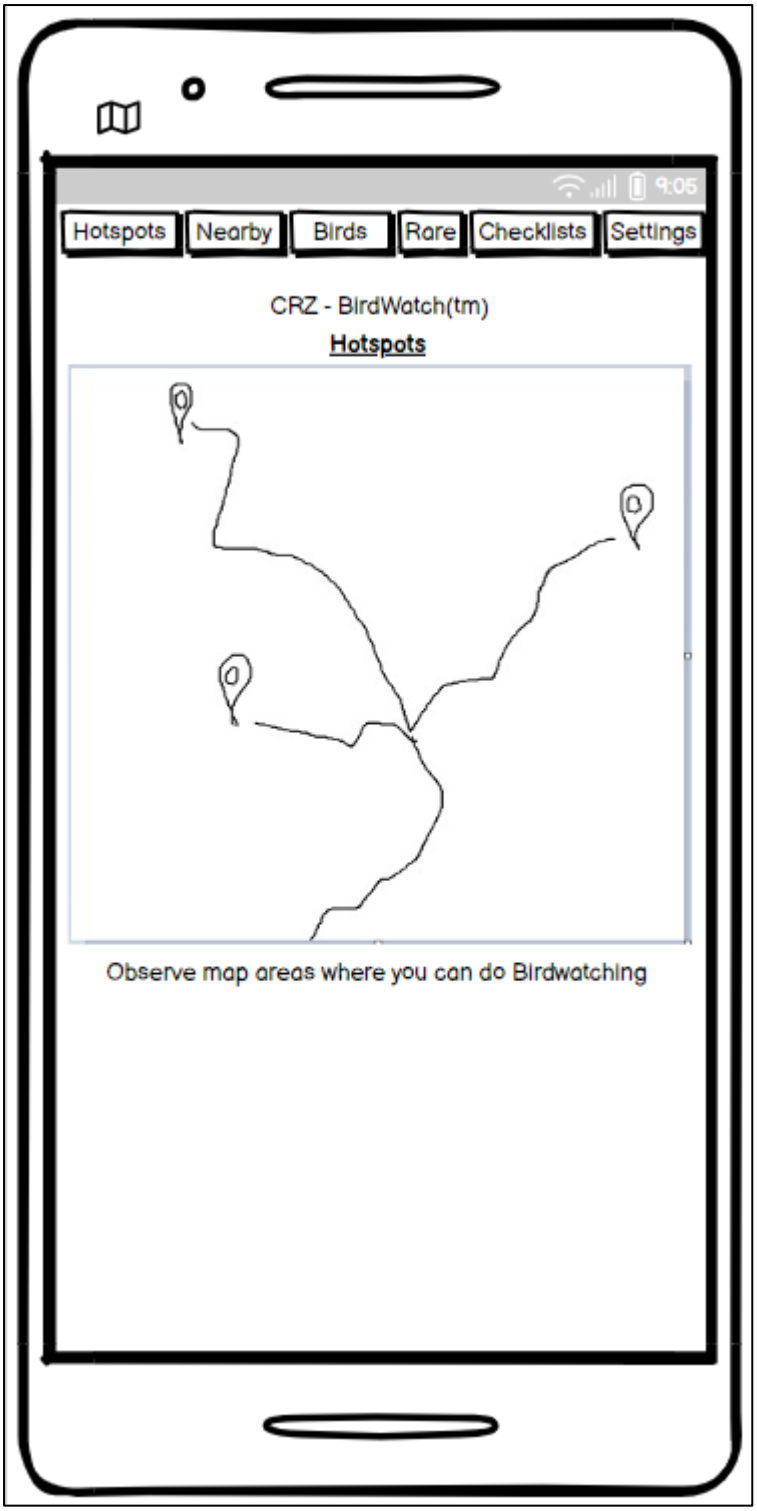


Figure 5

Hotspots Page

*'Observe map areas where you can do Birdwatching'.

This shows areas with bird activity.

This was sources from eBird by Cornell Lab (2023).

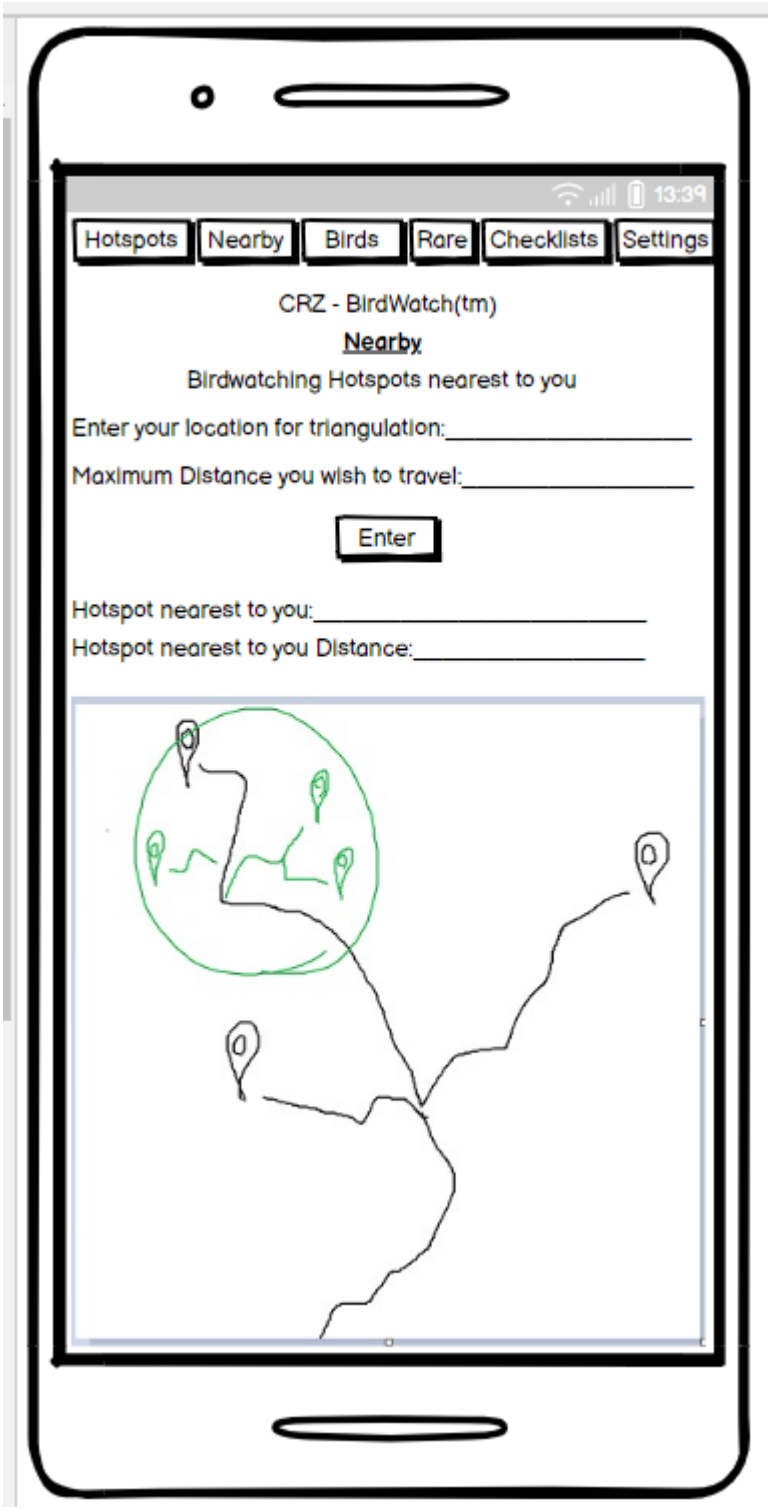


Figure 6

Nearby Page

*'Birdwatching Hotspots nearest to you'

This triangulates hotspots around your position. This was sourced from eBird by Cornell Lab (2023).

*'Hotspot nearest to you'

Shows the nearest possible hotspot

*'Hotspot nearest to you Distance:'

Calculates the distance from where you are to the Hotspot destination.

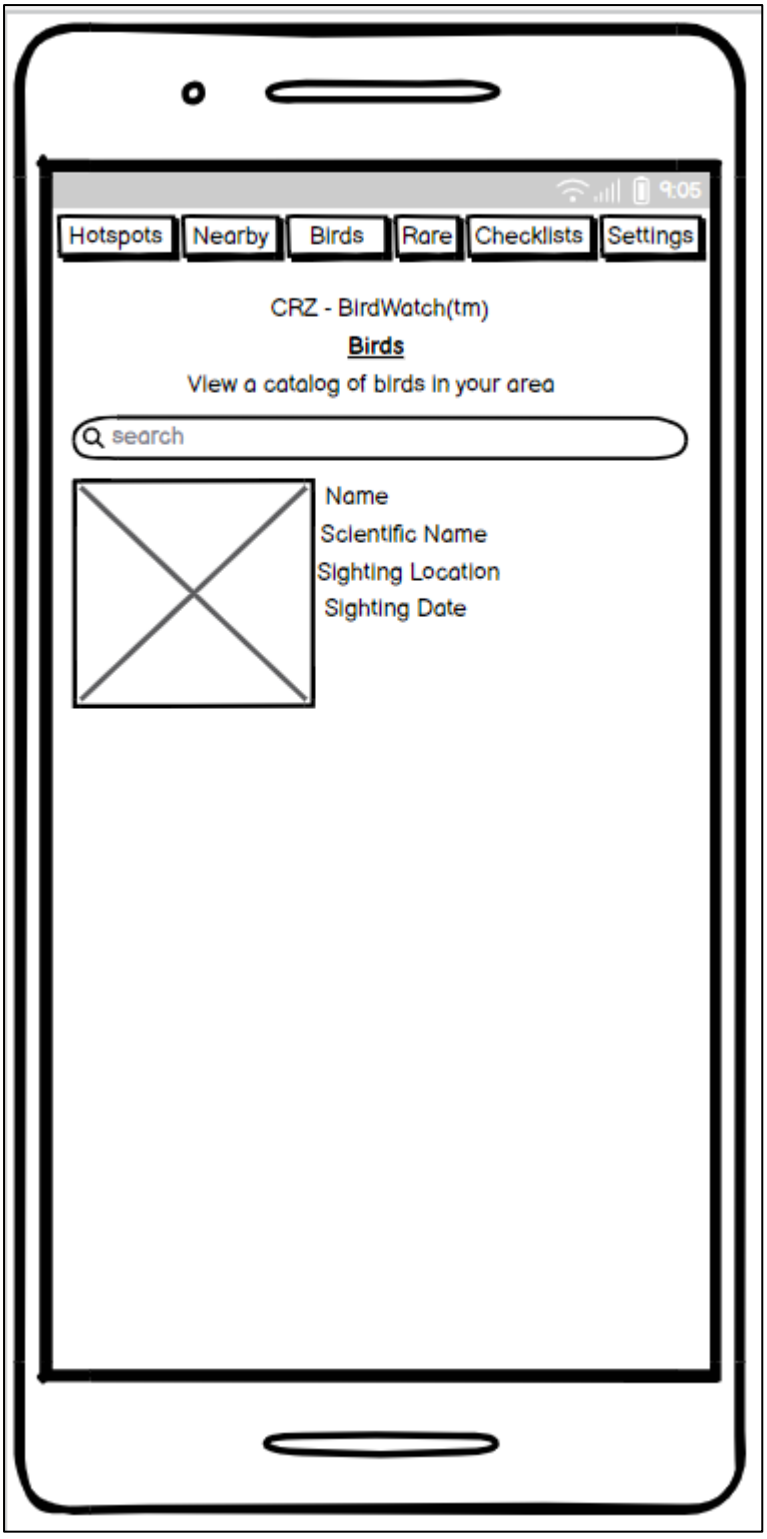


Figure 7

Birds Page

*'View a catalog of birds in your area'. View a database of birds and search for them.

Sourced from eBird by Cornell Lab (2023).

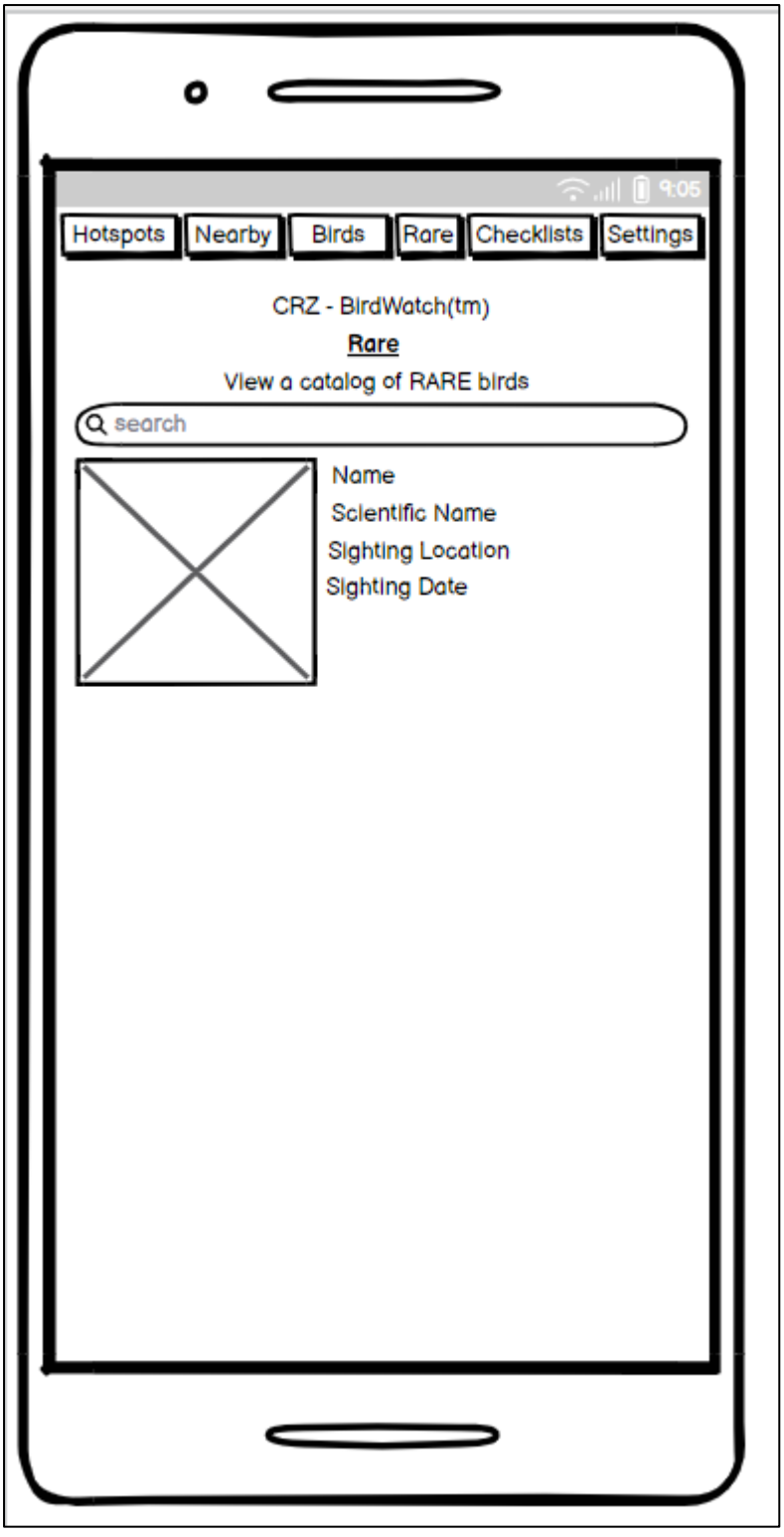


Figure 8

Rare Page

*'View a catalog of RARE birds'. View the rarest birds in the database and search their names.

Sourced from eBird by Cornell Lab (2023).



Figure 9

Checklist Page

This page allows Users to create and submit their bird watching checklists.

This system has been sourced from GoBird – Guide to Nearby Birds (2023).

*'NOT SUBMITTED'

These are checklists that have not been submitted. The space below the heading will display any checklist created not submitted.

*'SUBMITTED'

These are checklists that have been completed and added to the system. The space below the heading displays checklists that have been fully submitted.

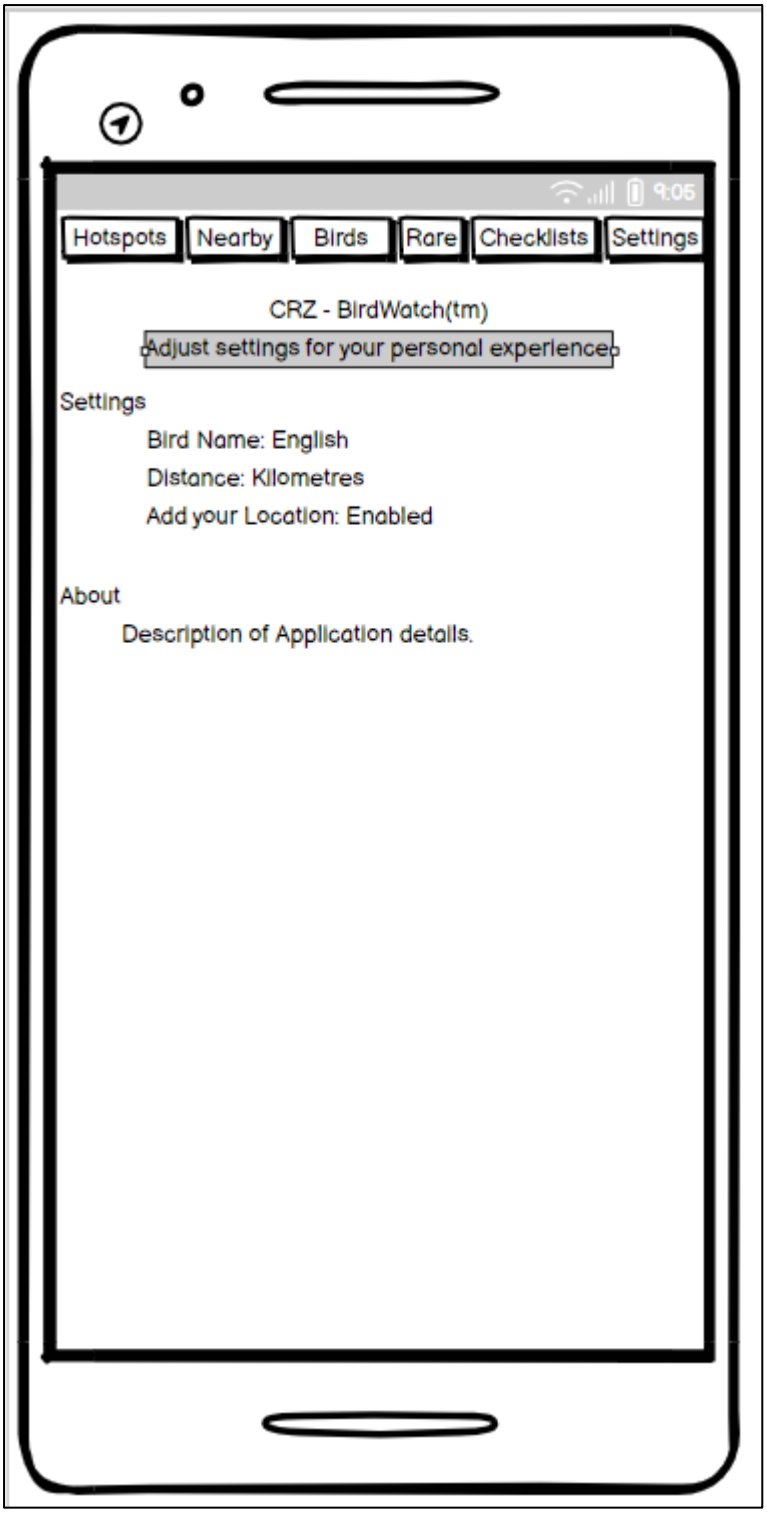


Figure 10

Settings Page

*'Adjust settings for your personal experience'. The User will be able to change settings on the App to better suit their own needs.

Obtained from GoBird – Guide to Nearby Birds (2023).

Here a User can:

*Change The language of the bird name.

*Change units of measurement for distances (such as kilometres or yards).

*Add their location, for the Map functions triangulation.

*View a description of the App's details.

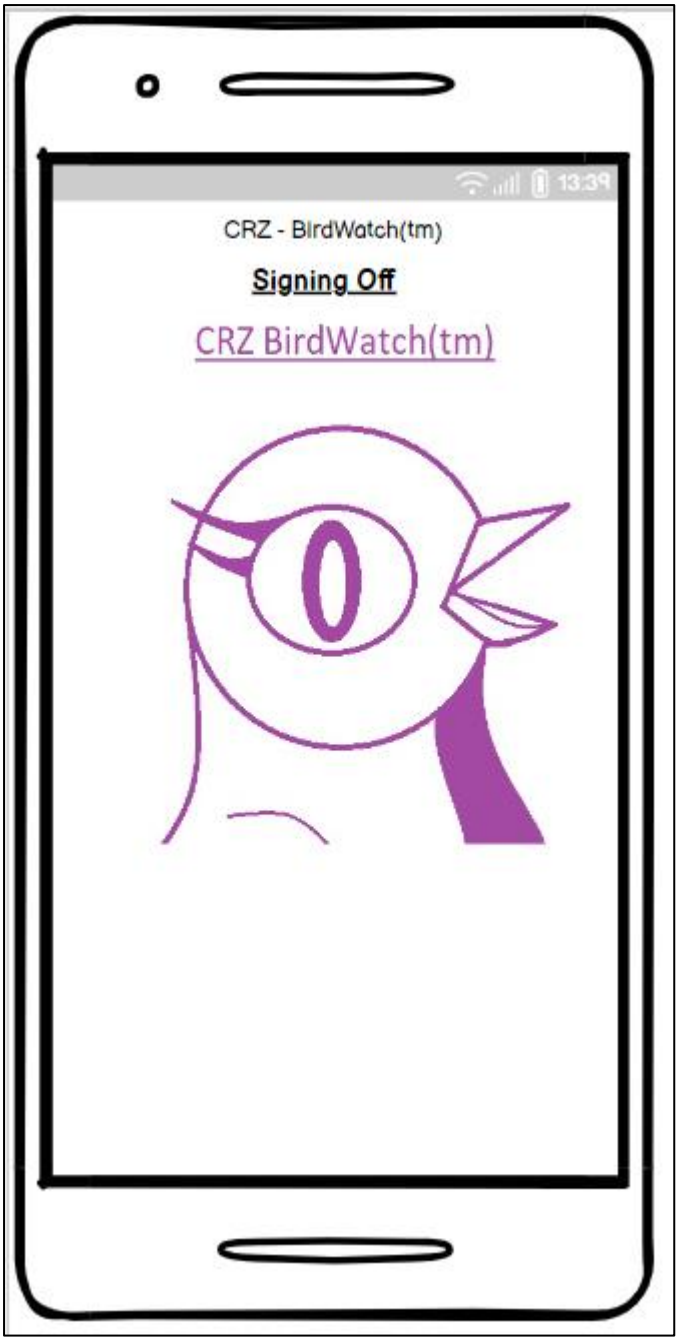


Figure 11

Exit Page

The page that appears as the App shuts down.

The App application will also, automatically 'sign out' the User's account, they properly enter back into the program each time they open it.

This page can't be manipulated. This practice was observed in Birda (2023), added for the sake of good branding.

Detailed listing of the data

Name (Web page)	Description/Explanation	Data Type(s)
Starting Page	The opening web page of the App.	
Sign up/Sign in Page	Create/access an account	1)Sign up Button - Input Data Type Format: Btn 2)Sign in Button - Input Data Type Format: Btn
Welcome Page	Page welcoming Users to App.	
Hotspots Page	Page displaying hotspots for bird activity.	
Nearby Page	Page displaying hotspots for bird activity, near users.	1)Your location: Physical location Data Type Format: Val 2)Enter Button Data Type Format: Btn 3)Map - Image File Data Type Format: Val
Birds Page	Catalogue/database of birds.	2)Bird Image - Image File Data Type Format:
Rare Page	Catalogue /database of birds, filtered by rareness.	1)Bird Image - Image File in App database. Data Type Format: Val
Checklists Page	Page to create, stores and submit checklists.	3)Date: Data Type Format: Val 4)Start time: Data Type Format: Val 5)NOT SUBMITTED: Data Type Format: Val 6)SUBMITTED: Data Type Format: Val
Settings Page	Where a user changes the settings.	1)Bird Name - Bird name Language. Data Type Format: Val 2)Distance - Distance Measurement. Data Type Format: Val 3)Add your location - User's physical position. Data Type Format: Val
Exit Page	Last web page as the App closes.	

The above table uses information from Ratabouil (2015) and Wilson (2016).

Project plan

The following images show a project plan created using MS Project (Microsoft, 2023).

The Overview:

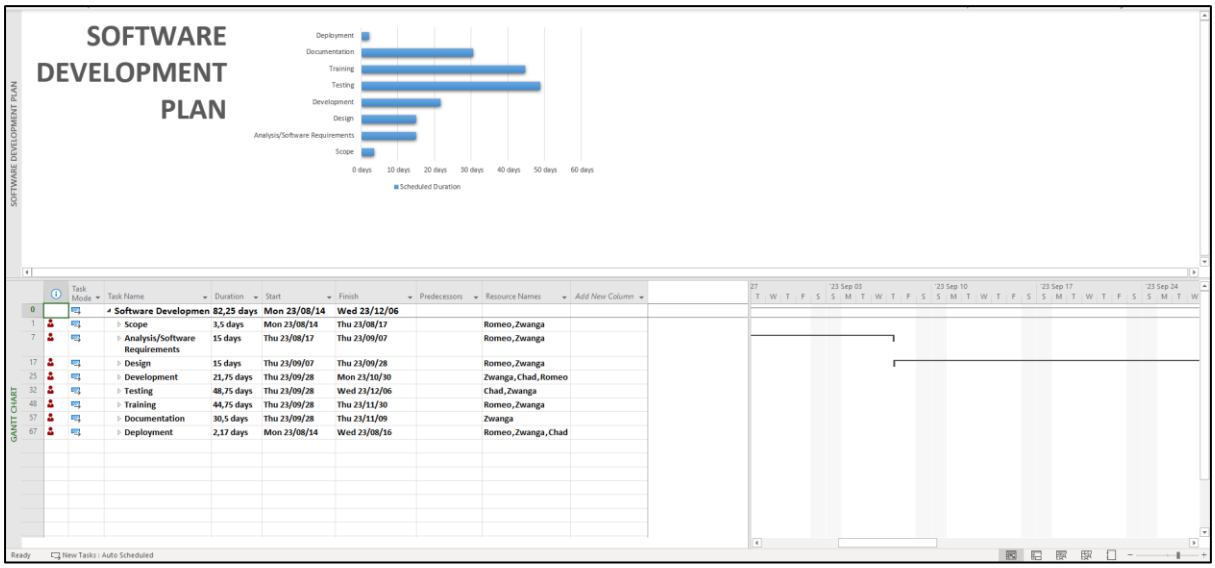


Figure 12

Additional screenshots:

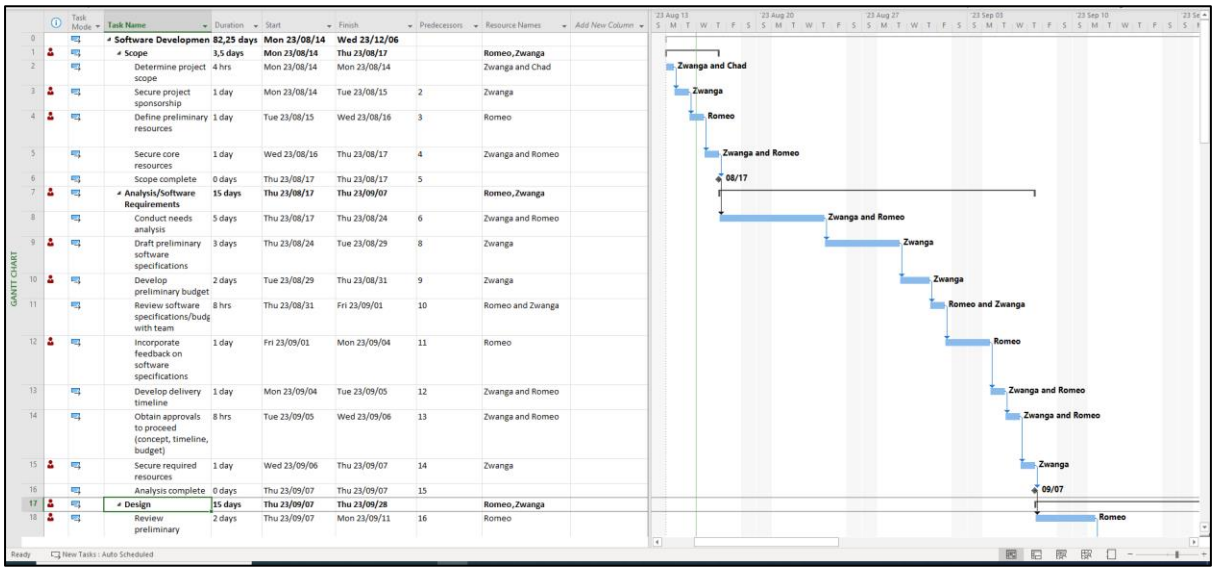
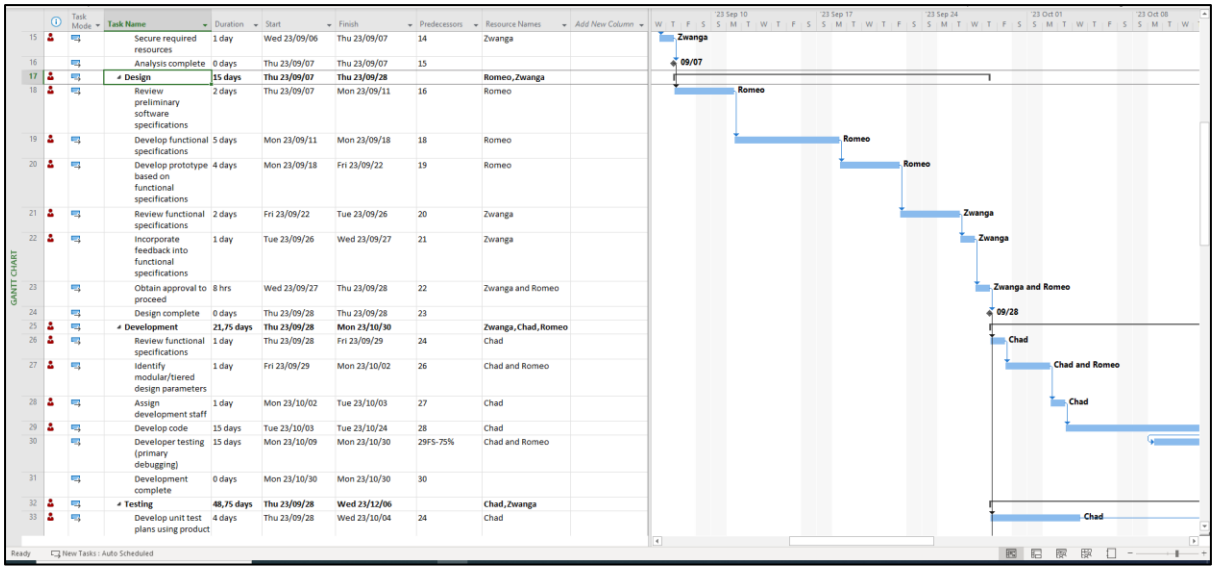
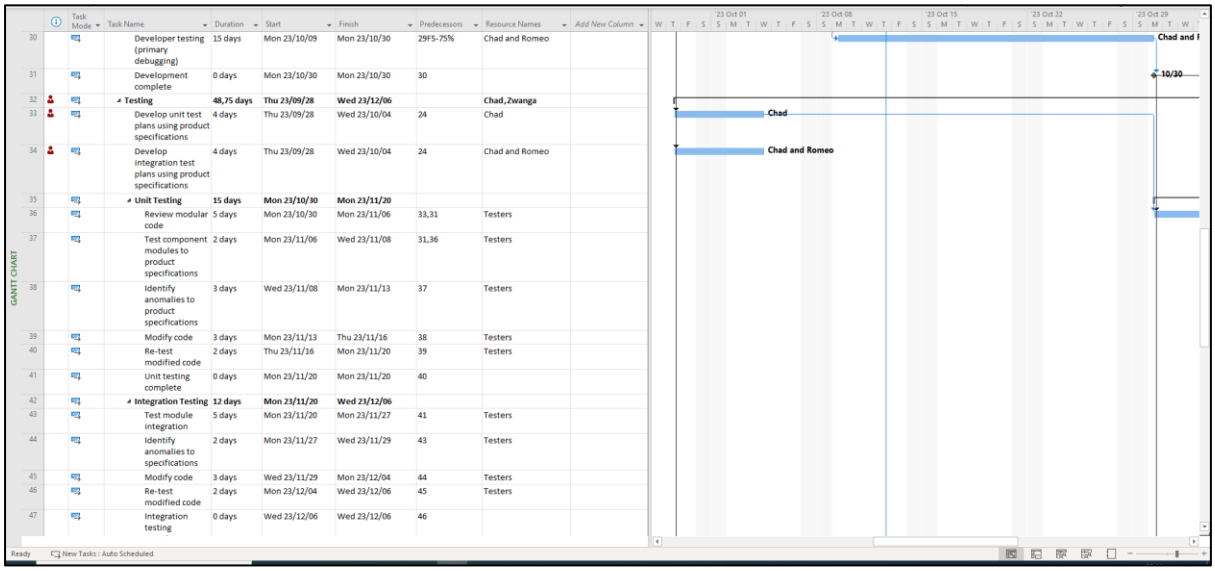


Figure 13



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
15		Secure required resources	1 day	Wed 23/09/06	Thu 23/09/07	14	Zwanga
16		Analysis complete	0 days	Thu 23/09/07	Thu 23/09/07	15	
17		**Design**	**15 days**	**Thu 23/09/07**	**Thu 23/09/28**		**Romeo,Zwanga**
18		Review preliminary software specifications	2 days	Thu 23/09/07	Mon 23/09/11	16	Romeo
19		Develop functional specifications	5 days	Mon 23/09/11	Mon 23/09/18	18	Romeo
20		Develop prototype based on functional specifications	4 days	Mon 23/09/18	Fri 23/09/22	19	Romeo
21		Review functional specifications	2 days	Fri 23/09/22	Tue 23/09/26	20	Zwanga
22		Incorporate feedback into functional specifications	1 day	Tue 23/09/26	Wed 23/09/27	21	Zwanga
23		Obtain approval to proceed	8 hrs	Wed 23/09/27	Thu 23/09/28	22	Zwanga and Romeo
24		Design complete	0 days	Thu 23/09/28	Thu 23/09/28	23	
25		**Development**	**21,75 days**	**Thu 23/09/28**	**Mon 23/10/30**		**Zwanga,Chad,Romeo**
26		Review functional specifications	1 day	Thu 23/09/28	Fri 23/09/29	24	Chad
27		Identify modular/tiered design parameters	1 day	Fri 23/09/29	Mon 23/10/02	26	Chad and Romeo
28		Assign development staff	1 day	Mon 23/10/02	Tue 23/10/03	27	Chad
29		Develop code	15 days	Tue 23/10/03	Tue 23/10/24	28	Chad
30		Developer testing (primary debugging)	15 days	Mon 23/10/09	Mon 23/10/30	29FS-75%	Chad and Romeo
31		Development complete	0 days	Mon 23/10/30	Mon 23/10/30	30	
32		**Testing**	**48,75 days**	**Thu 23/09/28**	**Wed 23/12/06**		**Chad,Zwanga**
33		Develop unit test plans using product	4 days	Thu 23/09/28	Wed 23/10/04	24	Chad

Figure 14



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
30		Developer testing (primary debugging)	15 days	Mon 23/10/09	Mon 23/10/30	29FS-75%	Chad and Romeo
31		Development complete	0 days	Mon 23/10/30	Mon 23/10/30	30	
32		**Testing**	**48,75 days**	**Thu 23/09/28**	**Wed 23/12/06**		**Chad,Zwanga**
33		Develop unit test plans using product specifications	4 days	Thu 23/09/28	Wed 23/10/04	24	Chad
34		Develop integration test plans using product specifications	4 days	Thu 23/09/28	Wed 23/10/04	24	Chad and Romeo
35		**Unit Testing**	**15 days**	**Mon 23/10/30**	**Mon 23/11/20**		
36		Review modular code	5 days	Mon 23/10/30	Mon 23/11/06	33,31	Testers
37		Test component modules to product specifications	2 days	Mon 23/11/06	Wed 23/11/08	31,36	Testers
38		Identify anomalies to product specifications	3 days	Wed 23/11/08	Mon 23/11/13	37	Testers
39		Modify code	3 days	Mon 23/11/13	Thu 23/11/16	38	Testers
40		Re-test modified code	2 days	Thu 23/11/16	Mon 23/11/20	39	Testers
41		Unit testing complete	0 days	Mon 23/11/20	Mon 23/11/20	40	
42		**Integration Testing**	**12 days**	**Mon 23/11/20**	**Wed 23/12/06**		
43		Test module integration	5 days	Mon 23/11/20	Mon 23/11/27	41	Testers
44		Identify anomalies to specifications	2 days	Mon 23/11/27	Wed 23/11/29	43	Testers
45		Modify code	3 days	Wed 23/11/29	Mon 23/12/04	44	Testers
46		Re-test modified code	2 days	Mon 23/12/04	Wed 23/12/06	45	Testers
47		Integration testing	0 days	Wed 23/12/06	Wed 23/12/06	46	

Figure 15

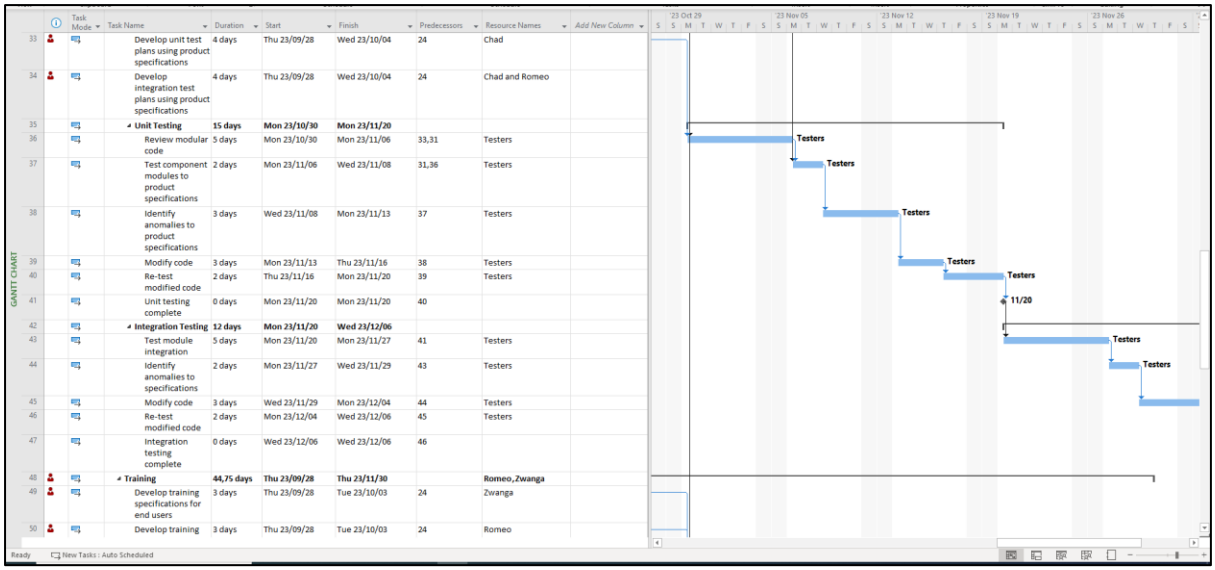


Figure 16

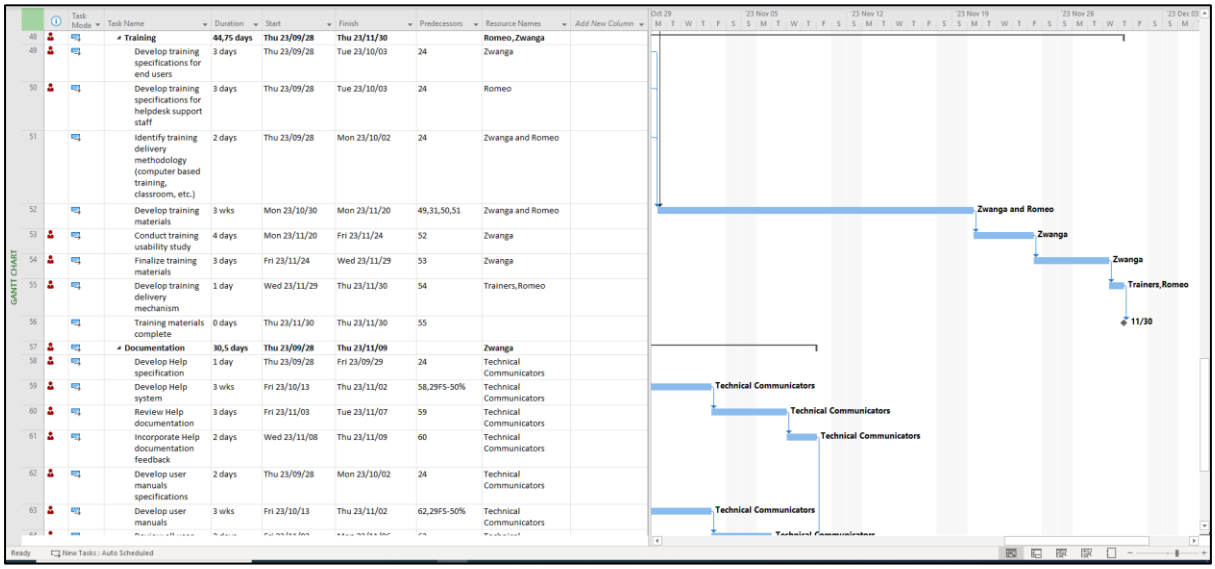
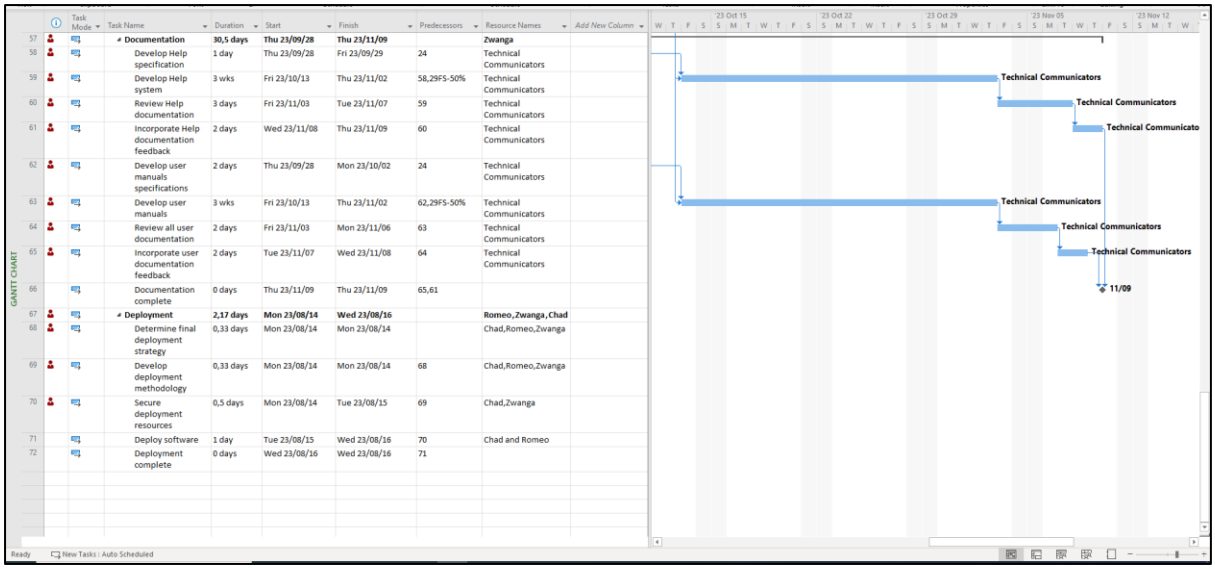


Figure 17



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
57		**Documentation**	**30,5 days**	**Thu 23/09/28**	**Thu 23/11/09**		**Zwanga**
58		Develop Help specification	1 day	Thu 23/09/28	Fri 23/09/29	24	Technical Communicators
59		Develop Help system	3 wks	Fri 23/10/13	Thu 23/11/02	58,29FS-50%	Technical Communicators
60		Review Help documentation	3 days	Fri 23/11/03	Tue 23/11/07	59	Technical Communicators
61		Incorporate Help documentation feedback	2 days	Wed 23/11/08	Thu 23/11/09	60	Technical Communicators
62		Develop user manuals specifications	2 days	Thu 23/09/28	Mon 23/10/02	24	Technical Communicators
63		Develop user manuals	3 wks	Fri 23/10/13	Thu 23/11/02	62,29FS-50%	Technical Communicators
64		Review all user documentation	2 days	Fri 23/11/03	Mon 23/11/06	63	Technical Communicators
65		Incorporate user documentation feedback	2 days	Tue 23/11/07	Wed 23/11/08	64	Technical Communicators
66		Documentation complete	0 days	Thu 23/11/09	Thu 23/11/09	65,61	
67		**Deployment**	**2,17 days**	**Mon 23/08/14**	**Wed 23/08/16**		**Romeo,Zwanga,Chad**
68		Determine final deployment strategy	0,33 days	Mon 23/08/14	Mon 23/08/14		Chad,Romeo,Zwanga
69		Develop deployment methodology	0,33 days	Mon 23/08/14	Mon 23/08/14	68	Chad,Romeo,Zwanga
70		Secure deployment resources	0,5 days	Mon 23/08/14	Tue 23/08/15	69	Chad,Zwanga
71		Deploy software	1 day	Tue 23/08/15	Wed 23/08/16	70	Chad and Romeo
72		Deployment complete	0 days	Wed 23/08/16	Wed 23/08/16	71	

Figure 18

Conclusion

As can be seen in the documentation here, the importance of *planning and design* dictates that these procedures must be carried out before any real programming can be done.

REFRENCE LIST

Birda. 2023. Birda: Birding made better. [App]. Available at: https://play.google.com/store/apps/details?id=com.chirpbirding.birda&hl=en_US [Accessed 8 August 2023]

Documenter.getpostman.com. 2023. eBird API 2.0. Available at: https://documenter.getpostman.com/view/664302/S1ENwy59#:~:text=The%202.0%20APIs%2are%20a,is%20JSON%20unless%20otherwise%20noted. [Accessed 20 August 2023]

eBird by Cornell Lab. 2023. Available at: https://play.google.com/store/apps/details?id=edu.cornell.birds.ebird&hl=en&gl=US [Accessed 8 August 2023]

GoBird – Guide to Nearby Birds. 2023. [App]. Available at: https://play.google.com/store/apps/details?id=com.thenerdbirder.GoBird&hl=en&gl=US [Accessed 8 August 2023]

GoogleMaps. 2023. GoogleMaps. [Online]. Available at: https://www.google.com/maps/@-33.9929343,25.5325795,15z?entry=ttu [Accessed 17 August 2023]

Microsoft. 2023. MS Project. [App]. Available at: Microsoft Project Download for Free - 2023 Latest Version (wizcase.com) [Accessed 17 August 2023]

Pypi.org. 2023. Project description. Available at: https://pypi.org/project/ebird api/#:~:text=eBird%20API%20provides%20a%20set,in%20the%20eBird%20API%202.0. [Accessed 20 August 2023]

Ratabouil, S. 2015. Android NDK: Beginner's Guide - Second Edition. [Online]. Available at: https://web-s-ebscohost-com.ezproxy.iielearn.ac.za/ehost/ebookviewer/ebook/ZTAwMHh3d19fOTg2MzczX19BTg2?sid=27fe9374-f654-4d04-9c81-b2be5cf3a87b%40redis&vid=5&format=EB&rid=1 [Accessed 20 April 2023]

The IIE. 2023. Open Source coding (Intemediate) 3A. [OPC7312POE]. The Independent Institute of Education: Unpublished.

Wilson, J. 2016. Creating Dynamic UIs with Android Fragments - Second Edition. [Online]. Available at: https://web-s-ebscohost-com.ezproxy.iielearn.ac.za/ehost/ebookviewer/ebook/ZTAwMHh3d19fMTIxMzUxMl9fQU41?sid=27fe9374-f654-4d04-9c81-b2be5cf3a87b%40redis&vid=1&format=EB&rid=1 [Accessed 20 April 2023]